DevOps 2TIN

Chapter 2

Agile, DevOps & the 3 ways

DE HOGESCHOOL MET HET NETWERK

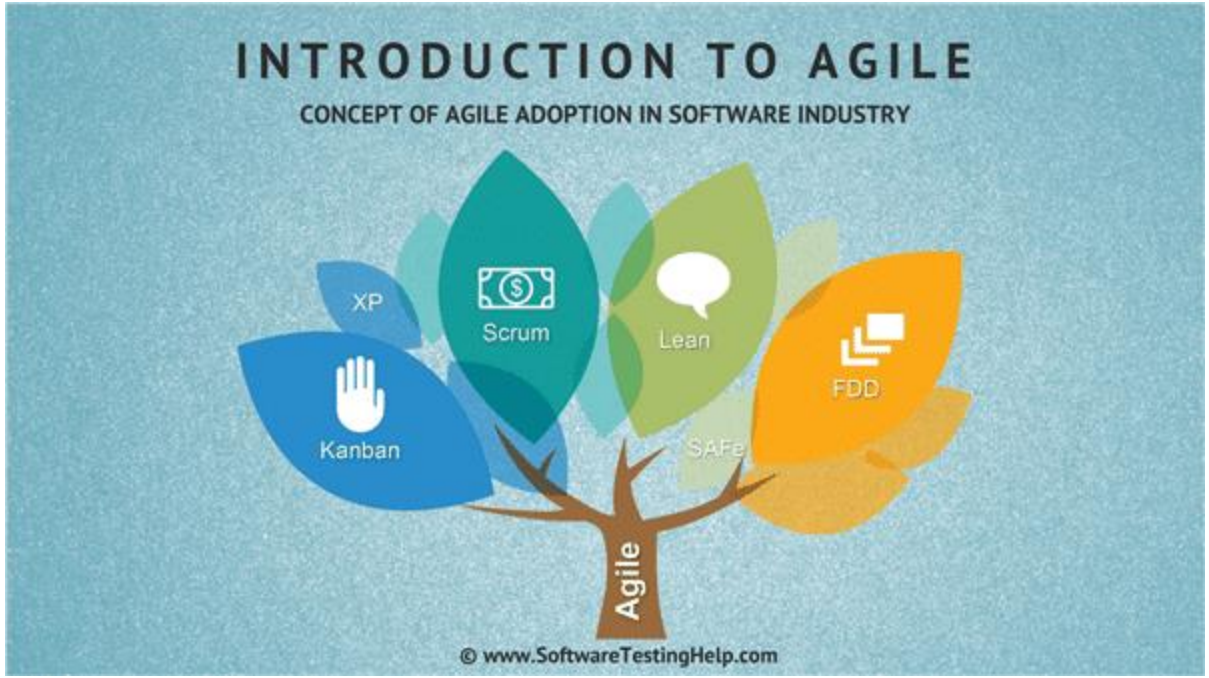
Elfde-Liniestraat 24, 3500 Hasselt, www.pxl.be

Agile, DevOps & the 3 ways

Agile vs DevOps
Historiek
The 3 ways
Reflectie

HOGESCHOOL PXL

Wat is agile?


INTRODUCTION TO AGILE
CONCEPT OF AGILE ADOPTION IN SOFTWARE INDUSTRY
XP
Scrum
Lean
FDD
Kanban
SAFe
Agile
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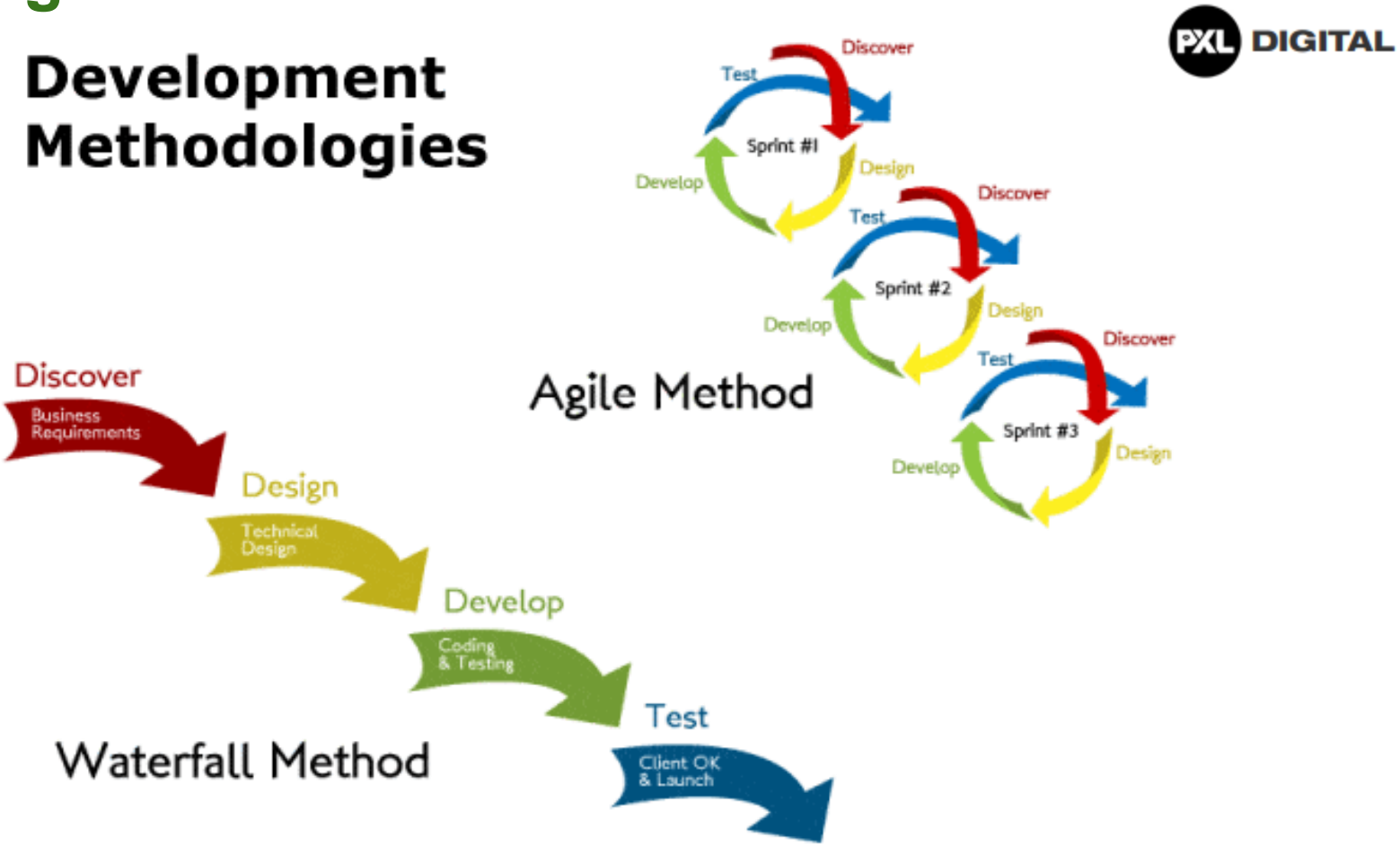
Kleine geschiedenis les



En dan nu DevOps?

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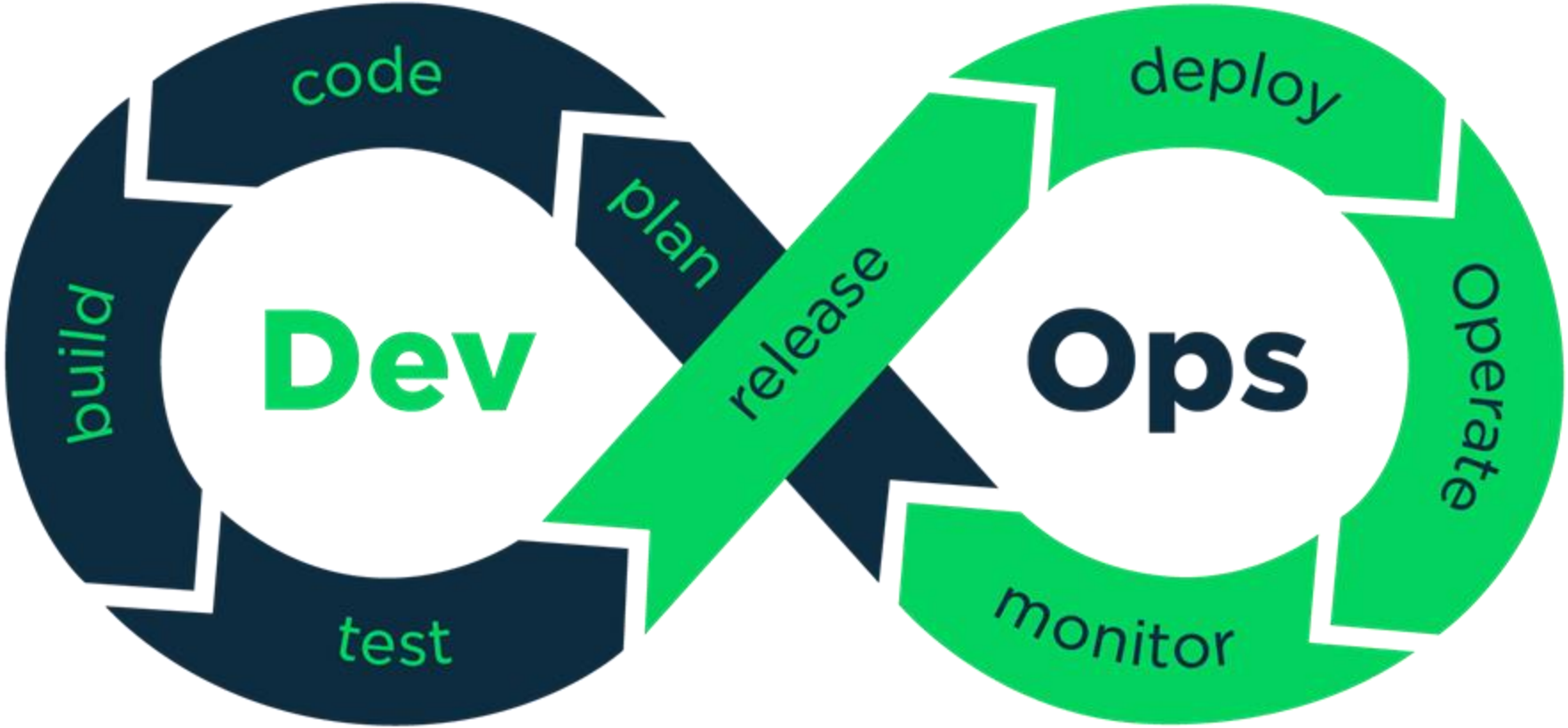

Dus DevOps is een praktische aanpak?

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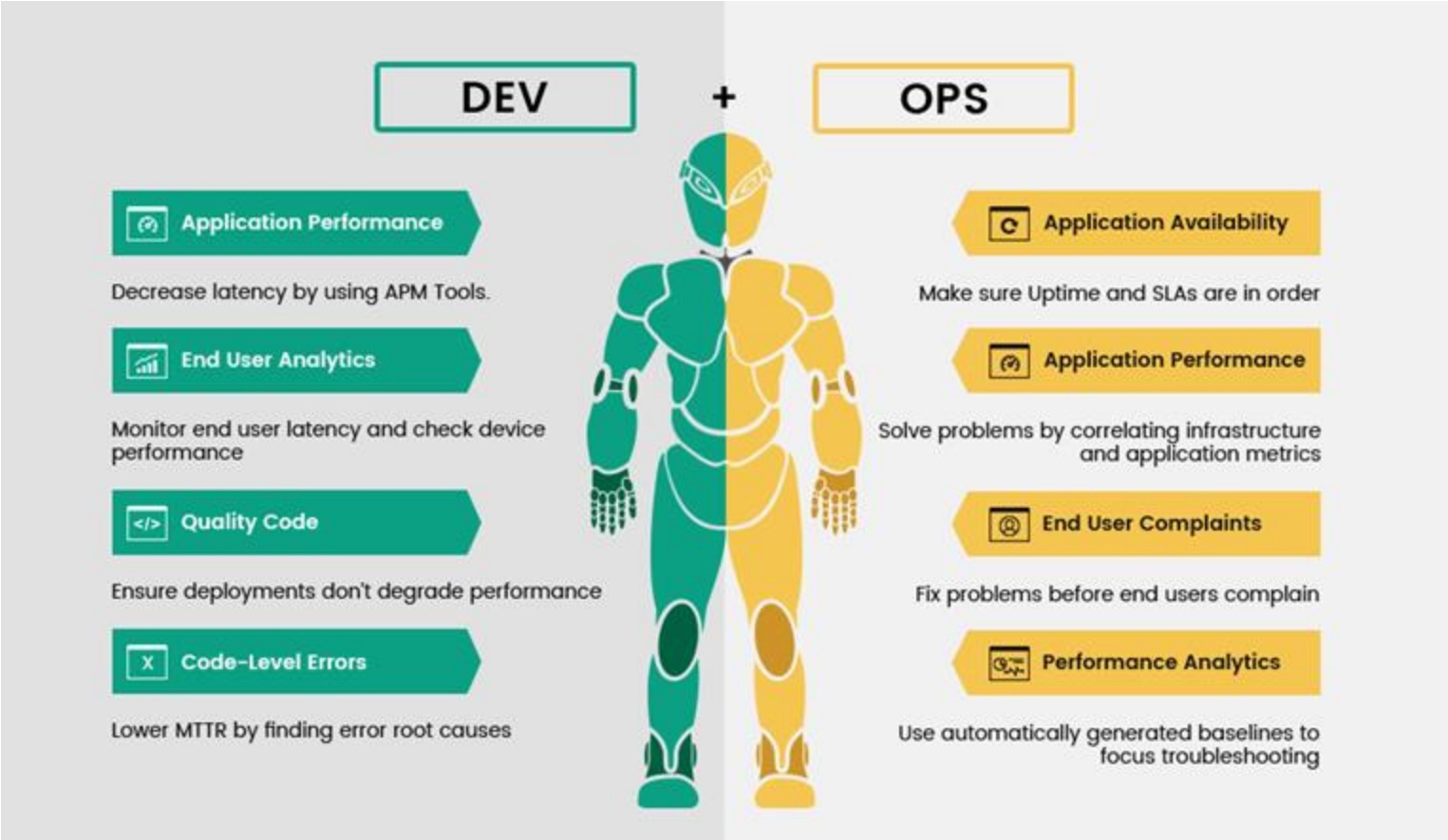
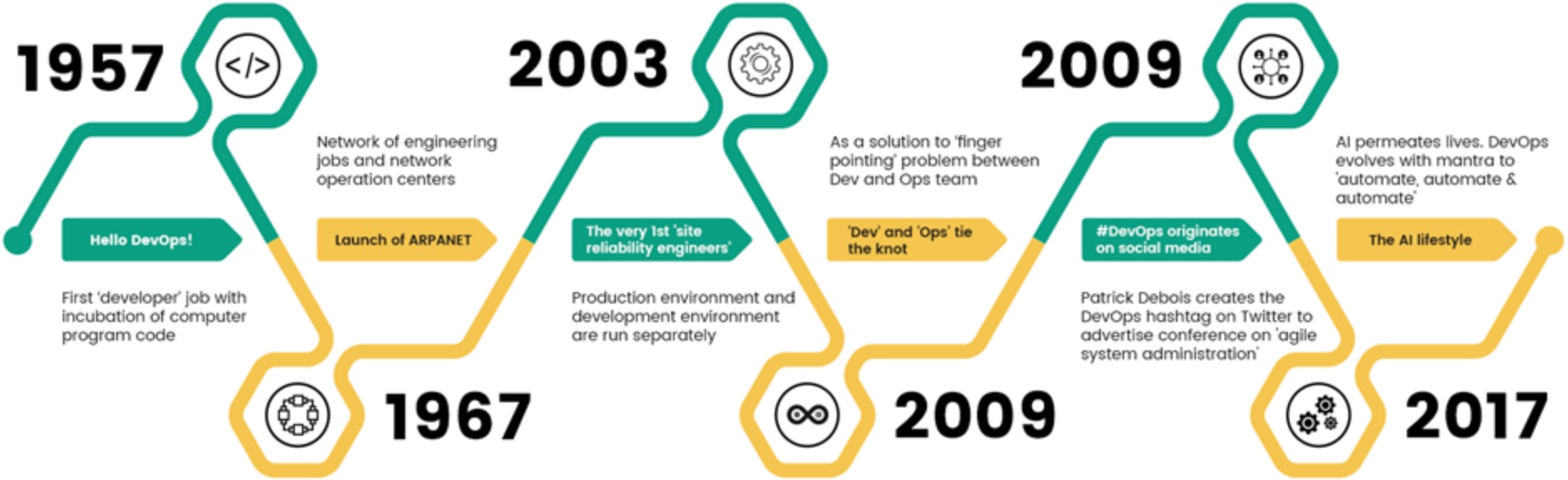


Ohja….DevOps is niet nieuw



source: https://dzone.com/articles/is-devops-agile

Veel dingen doen, maar waarom?

De Filosofie achter DevOps:

The 3 Ways

(of, waarom we al die dingen doen)

The first way

The principles of Flow

- ➔ Waterfall (van boven naar beneden)
- ➔ Agile (van boven naar beneden, in loopjes)
- ➔ DevOps (van links naar rechts naar links, altijd in beweging)

Maar, Hoe?

Hoe verhogen we de “flow”

Kleinere stapjes, maar meer releases betekent:

- Minder WiP
- Makkelijker om fouten te vinden en op te lossen
- Maakt werk zichtbaarder

Source: “Why mass production isn’t the most efficient way of doing ‘stuff’” by Stefan Luyten

The first way

Identificeer de onderdelen die zorgen dat het business-idee omgezet wordt in een realiteit

- Code, Infrastructuur, Data, Testen, Nieuwe features = Flow

En maak er een pipeline van ⟶ Eén systeem ivp verschillende silos

Wat is een Pipeline?

Definitie “deployment pipeline”:

(first defined by Jez Humble and David Farley in their book *Continuous Delivery: Reliable Software Releases Through Build, Test, and Deployment Automation*)

It ensures that all code checked in to version control is automatically built and tested in a production-like environment.

Sleutelwoorden:

- Alle Code
- Versiebeheer
- Automatisch gebouwd
- Automatisch getest
- Productie-waardige omgeving

THE FIRST WAY

CREATE A DELIVERY SYSTEM

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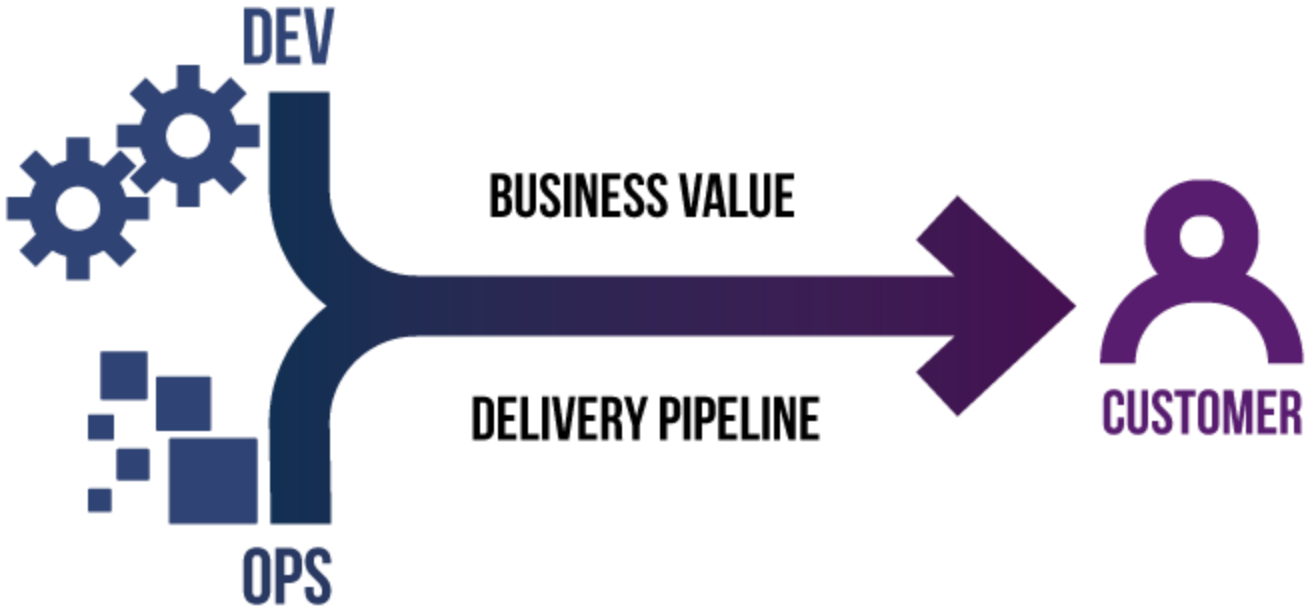


The Second Way

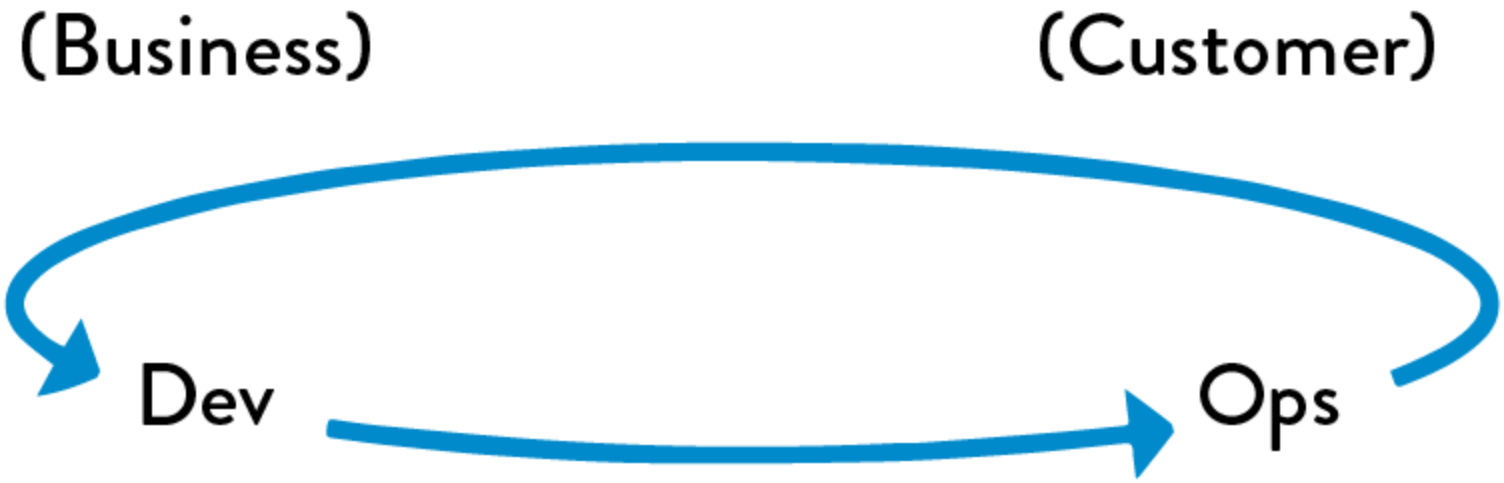
The principles of feedback

- ➔ Waterfall (enkel op het einde)
- ➔ Agile (enkel op het einde van de sprint)
- ➔ DevOps (continue)

Maar, Hoe?

The Second Way

Door continue te monitoren & testen en ervoor te zorgen dat we problemen opmerken als ze gebeuren, ze aan te pakken en op te lossen alvorens verder te gaan



Test **ALLES** vanaf het moment dat het kan!

Test **automatisch** om Continous Integration te bekomen

Definitie *continuous integration*:

- **A comprehensive and reliable set of automated tests that validate we are in a deployable state.**
- **A culture that "stops the entire production line" when our validation tests fail. (=digital andon cord)**

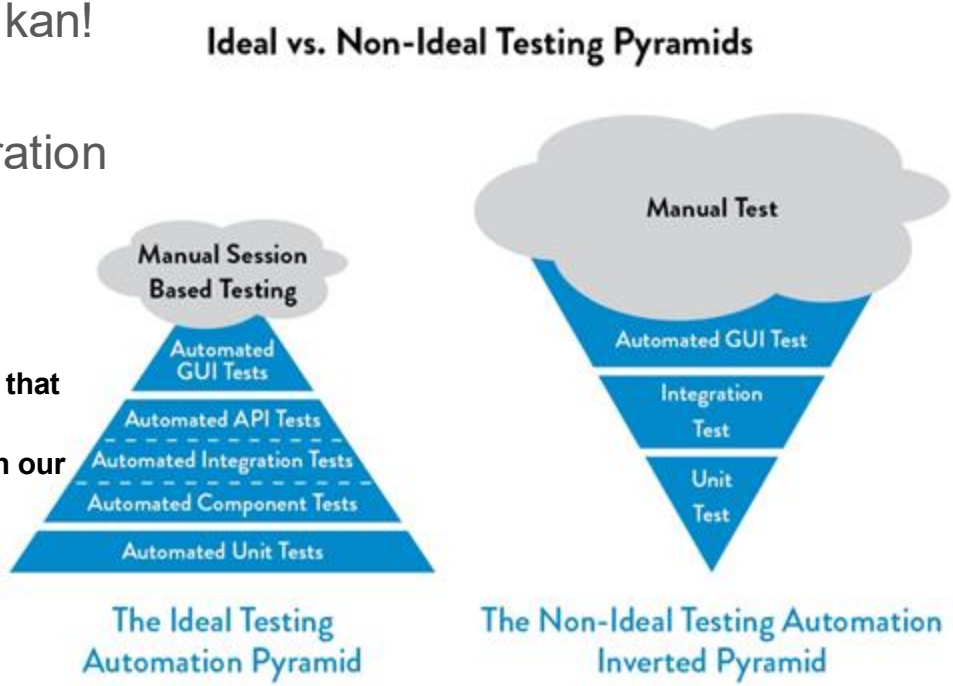


Figure 14: *The ideal and non-ideal automated testing pyramids (Source: Martin Fowler, "TestPyramid.")*

Stop de pipeline, gooi het probleem in de groep en los het op
Swarming zorgt ervoor dat dit snel en efficiënt kan

Source: https://medium.com/devops-cloud-it-career/the-three-ways-of-devops-part-2-of-3-8d473f50d99d

Problemen NU oplossen voorkomt issues later!

Problemen laten liggen “omdat er nu geen tijd is” zorgt voor **Technical Debt**

Source: https://medium.com/devops-cloud-it-career/the-three-ways-of-devops-part-2-of-3-8d473f50d99d

THE SECOND WAY

AMPLIFY FEEDBACK LOOPS

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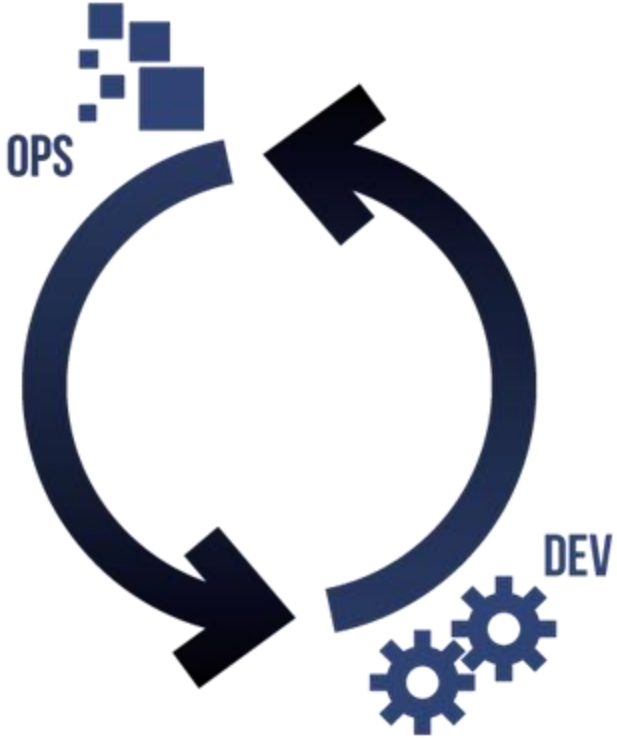


The Third Way

the creation of a generative, high-trust culture that supports a dynamic, disciplined, and scientific approach to experimentation and risk-taking, facilitating the creation of organizational learning, both from our successes and failures

Enable safety

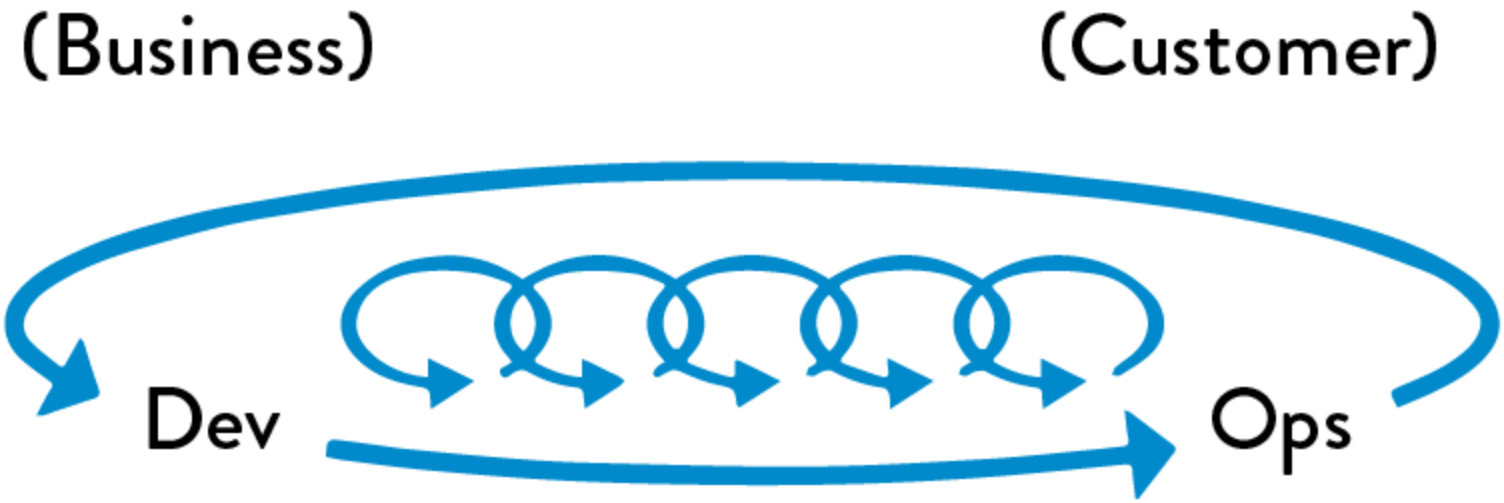
Improvement of daily work

Local discoveries to global improvements

Inject Resilience

The Third Way

Dit is de moeilijkste, en heeft vaak ook te maken met een cultuur shift binnen bedrijven.



THE THIRD WAY

LEARN & EXPERIMENT CONTINUOUSLY

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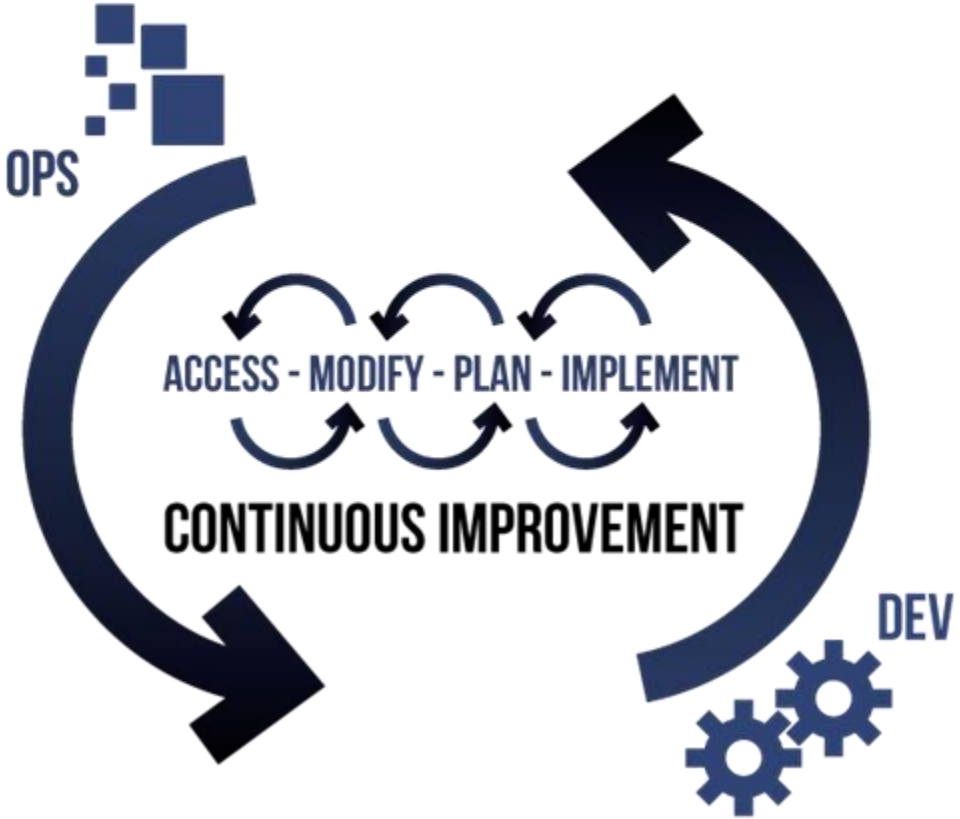
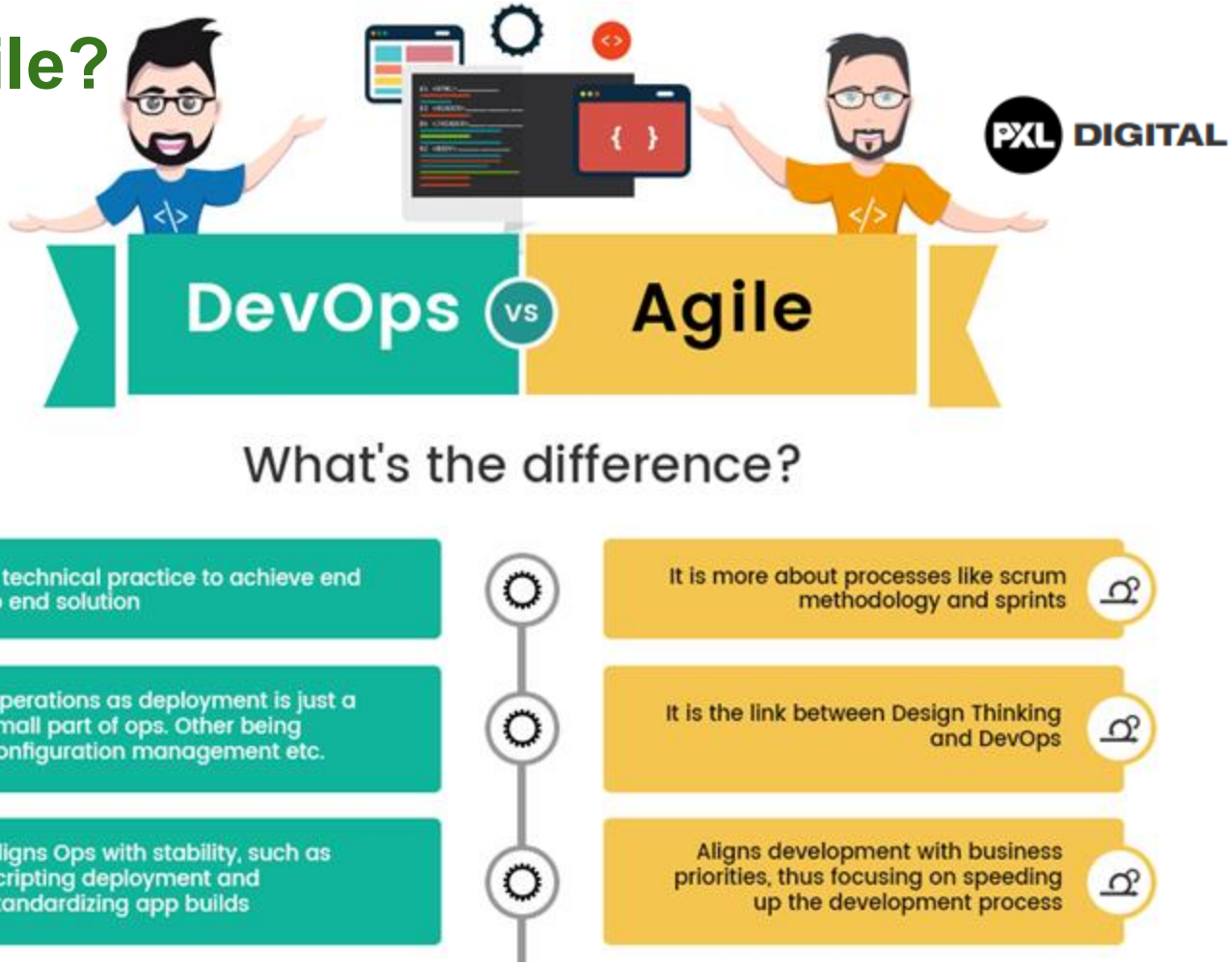


High level overview, hoe we dit in de praktijk gaan doen zien we in de volgende hoofdstukken vanuit de praktijk.

Is DevOps Agile?



source: https://dzone.com/articles/is-devops-agile

Recap - Is DevOps Agile?

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DevOps helps to bridge the line between several cross functional teams (Development, QA, Monitoring, Deployment, Infrastructure etc.) by simplifying the process

In agile, the gap between these teams are bridged in daily scrum calls where all roadblocks are discussed and resolved with agile scrum methodologies.

Aims at faster deliveries through automation via various applications and tools like Docker, Chef, Jira, Configuration Management and Splunk. These tools are used for automating deployment and tracking bugs in real-time for quick fix.

Aims at streamlined delivery via tools like Kanban Board, Scrum, Extreme Programming (XP), Test Driven Development (TDD), Feature Driven Development (FDD), and Lean. These practices help in project management, planning and collaboration.

source: https://dzone.com/articles/is-devops-agile